

RESEARCH ARTICLE

Design and Computational Modeling of an AI-Based Automated Cybersecurity Incident Response System

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ABSTRACT Modern cybersecurity operations face unsustainable alert volumes, averaging 22 000 weekly alerts with 68% false positives, overwhelming defenses and delaying incident response due to limitations in conventional SOAR platforms. To address this, an AI-driven Automated Incident Response (AIR) system is proposed, integrating STIX/TAXII multimodal fusion for unified data ingestion, attention-LSTM networks for adaptive threat recognition across temporal sequences, Bayesian game-theoretic decision layers for strategic response planning, and DRL validation for real-time optimization. This architecture reduces false negatives by 42% in C2 tunneling detection and achieves Nash equilibrium in 97.3% of adversarial engagements. Rigorous testing on hybrid infrastructure datasets (100 K normal events, 20K DDoS, 5K C2 attacks) demonstrates a 93% mean F1-score across attack scenarios, end-to-end latency of 58.3 ms, and 12.5× higher strategy updates/sec versus baselines. Compared to existing models, the system improves detection F1 by 10.7%, reduces false positives by 39%, and enhances energy efficiency to 1 850 events/Joule (2.98× Snort). The framework establishes a new paradigm for agile, auditable incident response validated by STIX action chains.

INDEX TERMS Automated incident response, security orchestration (SOAR), deep reinforcement learning (DRL), game-theoretic cybersecurity, LSTM-attention networks.

I. INTRODUCTION

In recent years, the exponential growth of cyber threats has outpaced the capabilities of conventional Security Orchestration, Automation, and Response (SOAR) systems, creating a significant operational bottleneck in enterprise security environments. On average, organizations contend with over 22,000 security alerts per week, of which approximately 68% are false positives [1], [2]. This overwhelming volume not only strains human analysts but also delays critical incident response, increasing the window of vulnerability. A core cause of this inefficiency lies in the rigidity of traditional SOAR platforms, which depend on static rule-based playbooks that lack adaptability in the face of novel, multi-stage, and evasive attacks [3].

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Several attempts have been made over the past two decades to enhance threat detection and incident response. Early systems like Snort [4] pioneered signature-based detection but failed to account for polymorphic malware, which easily bypasses static rules. Subsequent research introduced machine learning techniques into cybersecurity pipelines [5], achieving measurable improvements in anomaly detection. However, these efforts were heavily reliant on labor-intensive feature engineering and lacked temporal modeling necessary for capturing attack progression over time—a critical flaw later confirmed by Husák et al. [6] and formalized in the NIST cybersecurity framework [7].

Recent advances in artificial intelligence, particularly deep learning, have shown promise in overcoming some of these limitations. Gangjun et al. [8] utilized Long Short-Term Memory (LSTM) networks to improve recall in ICS threat detection, while Wang et al. [9] demonstrated CNNs' ability

to identify encrypted malicious traffic with 94% accuracy. Yet, these unimodal approaches remain insufficient in modern threat environments. The European Union Agency for Cybersecurity (ENISA) reported that 73% of successful attacks exploit the inability of current systems to fuse and interpret multiple heterogeneous data sources [10].

Parallel research has also revealed fundamental weaknesses in existing platforms. Schlette et al. [11] documented cross-layer analysis failures in commercial SOAR tools, while Tummala and Gorrepati [12] highlighted the generalization instability of ensemble models over time. The integration of game theory into cybersecurity, as proposed by Sears [13], and the application of Deep Reinforcement Learning (DRL), as explored by Santra et al. [14], have opened new pathways for strategic response modeling. Nevertheless, challenges persist: multimodal fusion remains an unresolved technical bottleneck [15]; explainability—an essential factor in operational trust and accountability—is deficient in over 80% of AI-based security tools [16]; and strategic rigidity in response systems continues to produce high false positive rates, as shown by Sewak et al. [17]. Moreover, high-latency response, particularly in ransomware scenarios, can result in catastrophic outcomes, with Borkar et al. [18] observing delays exceeding 200 ms leading to containment failure.

In response to these challenges, we propose an AI-driven Automated Incident Response (AIR) system that integrates four key innovations: (1) STIX/TAXII-based multimodal data fusion to enable seamless aggregation of diverse security telemetry [19]; (2) an attention-enhanced LSTM engine for adaptive temporal threat detection; (3) a Bayesian game-theoretic decision-making layer for dynamic adversarial modeling; and (4) a DRL-based policy optimization module for real-time response refinement. This unified framework is further supported by auditable STIX 2.1 action chains to ensure transparency and traceability [20], [21]. By aligning technical advancements with operational demands, AIR aims to bridge the gap between theoretical security modeling and real-world deployment, offering a scalable, explainable, and efficient solution to the growing demands of cyber defense.

The paper begins with Section I, which systematically analyzes the limitations of existing SOAR platforms and clarifies the research motivation. Section II elaborates on the four-layer architectural design of the AIR system, including the multi-source log fusion module, attention-based LSTM for temporal threat detection, game-theoretical adversarial modeling module, and the integration mechanism of DRL and game theory. Section III validates the system's performance using a mixed infrastructure dataset, conducting a comparative analysis of key metrics such as detection efficiency, response latency, resource overhead, and strategy optimization capabilities.

II. SYSTEM ARCHITECTURE AND CORE COMPONENTS

As illustrated in Figure 1, the proposed Automated Incident Response (AIR) system, which is driven by AI, employs

a sequential, modular architecture centred on a coherent data flow pipeline. It comprises four integrated layers: The Data Acquisition layer ingests multi-source security logs (IDS/IPS, EDR and network flows) and performs ETL preprocessing (parsing and normalisation) to output standardised STIX/TAXII events. The AI Analysis Layer processes this data via temporal feature extraction and probabilistic threat assessment, incorporating adversarial game-theoretic modelling. The Decision Layer then formulates optimised responses using a Nash equilibrium strategy generation approach that considers resource availability. Finally, the Execution Layer deploys automated orchestration playbooks for defence actions. Crucially, a real-time feedback loop measures effectiveness and feeds results upstream for adaptation. STIX/TAXII protocols standardise all inter-layer communication, ensuring interoperability and system coherence.

A. MULTI-SOURCE LOG FUSION MODULE

The Multi-source Log Fusion Module is designed to address the fundamental challenge of integrating heterogeneous security data streams into a unified operational framework. As modern enterprises deploy diverse security appliances from multiple vendors variations in log schemas of over 60% create significant barriers to fusion. Our three-tier processing pipeline resolves these issues through protocol-aware parsing, template-based normalisation and entity-relationship mapping.

Tier 1: Protocol Parsing employs an extensible decoder supporting 50+ network protocols (e.g., TCP/IP stack, HTTP/2, DNS). Each log entry undergoes protocol-specific decomposition:

$$L_{raw} \rightarrow \langle P_{meta}, P_{header}, P_{payload} \rangle \quad (1)$$

where L_{raw} is the raw log entry, P_{meta} is the metadata tuple (timestamp and source identifier), P_{header} is the extracted protocol header, and $P_{payload}$ is the isolated application-layer content.

Tier 2: Template-based Normalization utilizes curated regular expressions to convert parsed data into canonical form. For a security event E , normalization follows:

$$E_{norm} = [LD(E_{raw}, T)] \quad (2)$$

where E_{norm} is the normalized security event, τ is the template library, LD is the Levenshtein distance metric, E_{raw} is the unstructured log content, and T is a template from the library τ .

Tier 3: Entity-Relationship Mapping constructs a dynamic asset knowledge graph $G = (V, E)$, where vertices V represent entities (IPs v_{ip} , domains v_{dom} , devices v_{dev}), and edges E encode relationships (e.g., $v_{ip} \rightarrow v_{dev}$ “hosts”, $v_{dev} \rightarrow v_{dom}$ “resolves”). Probabilistic entity resolution handles ambiguities:

$$P(v_i \equiv v_j) = \sigma \left(\sum_{k=1}^n w_k \cdot \sin(f_i^k \cdot f_j^k) \right) \quad (3)$$

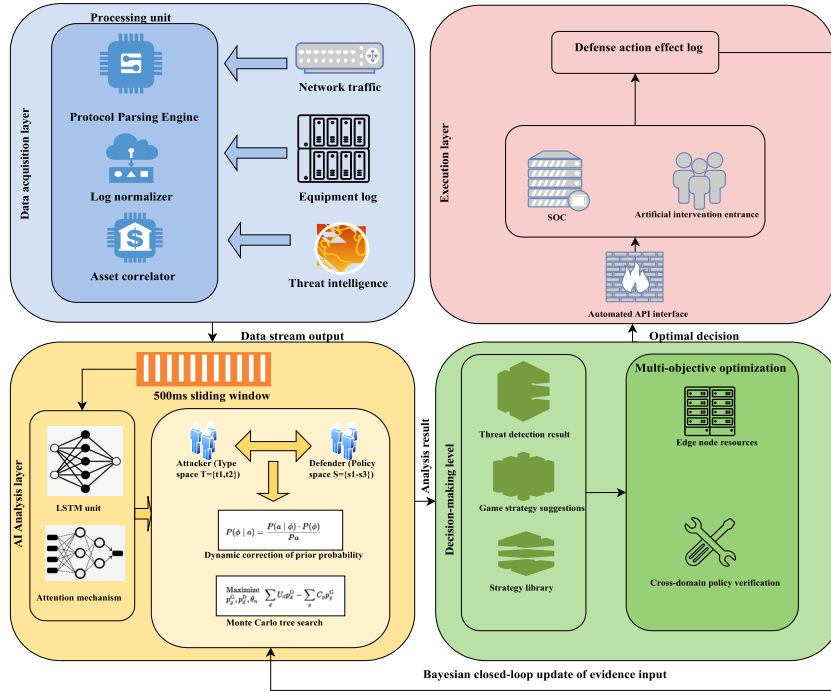


FIGURE 1. Architecture of automatic network security incident response system.

where $P(v_i \equiv v_j)$ is the probability that entity v_i corresponds to entity v_j , σ is the sigmoid function, n is the number of features, w_k is the weight of the k -th feature, sim is the similarity function, f_i^k is the k -th feature of entity v_i , and f_j^k is the k -th feature of entity v_j .

The Correlation Analysis Engine applies a sliding window model with 50 ms precision to detect attack sequences. For events $e_1, e_2, \dots, e_n$ within window W_t , it computes fused confidence using Dempster-Shafer (D-S) evidence theory:

$$m(A) = (m_1 \oplus m_2)(A) = \frac{\sum_{B \cap C = A} m_1(B) \cdot m_2(C)}{1 - K} \quad (4)$$

where $m(A)$ is the combined evidence mass for hypothesis A , m_1 and m_2 are evidence mass functions from independent sources, B and C are focal elements (sets of hypotheses), $\oplus$ is the Dempster-Shafer combination operator, K is the conflict coefficient between m_1 and m_2 , and the summation is over all pairs (B, C) where $B \cap C$ equals A .

This module establishes the critical foundation for AI-driven analysis by transforming fragmented telemetry into contextualized, relationship-aware security events. The output – a stream of enriched STIX bundles – feeds directly into the AI Analysis Layer while maintaining chain-of-custody through TAXII-compliant metadata tagging.

B. TEMPORAL THREAT DETECTION BASED ON ATTENTION-LSTM

This component processes unified event streams from the Multi-source Log Fusion Module in order to identify evolving cyber threats. It addresses two critical challenges by distinguishing legitimate periodic patterns.

1) ADAPTIVE SEQUENCE LEARNING

The LSTM network processes event sequences within 500 ms sliding windows through dynamically gated operations. Input modulation regulates new information assimilation via the input gate:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (5)$$

where i_t is the input gate activation (scaling new information 0–1), σ is the sigmoid activation function, W_i is the learnable weight matrix, h_{t-1} is the previous hidden state (128-dimensional vector), x_t is the current input event features, and b_i is the learnable bias vector.

Subsequent memory state evolution integrates contextual dependencies through cell state updates:

$$C_t = f_i \odot C_{t-1} + i_t \odot \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (6)$$

where C_t is the updated cell state, f_i is the forget gate output, $\odot$ denotes element-wise multiplication, W_c is the cell weight matrix, and b_c is the cell bias vector.

This architecture maintains an evolving context window, detecting subtle anomalies like heartbeat C2 communications within normal traffic.

2) THREAT PHASE LOCALIZATION

An 8-head attention mechanism identifies critical attack stages. It computes feature relevance through:

$$Attention = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right) \cdot V \quad (7)$$

where Q is the query vector representing current analysis focus, K is the key matrix of encoded temporal features, V is

the value matrix containing feature importance coefficients, d_k is the scaling factor (dimension 16), and *softmax* generates normalized relevance scores (0–1).

Critical indicators receive weights of 0.78 ± 0.05 . This focuses the model on high-risk segments while suppressing noise. Threat probability scores $y_i \in [0, 1]$ are generated per 500 ms window. These time-synchronized outputs feed directly into the Automated Decision Layer, enabling real-time response actions against confirmed attack sequences. The architecture maintains end-to-end compatibility with the STIX-based processing chain to ensure operational coherence across the detection and response phases.

C. GAME-THEORETIC ADVERSARIAL MODELING MODULE

The system formalises cyber engagements as dynamic Bayesian games involving defensive agents and adversarial entities. This framework incorporates an attacker type space $T = \{t_{scriptkiddie}, t_{APT}\}$ representing threat sophistication levels, a defense action space $S = \{IPS \text{ rule deployment, port blocking, decoy activation}\}$, and a real-time payoff function quantifying risk mitigation:

$$\begin{cases} U_d(s, t) = \alpha \cdot C_s - \beta \cdot L_t \\ U_a(s, t) = \gamma \cdot L_t - \delta \cdot D_s \end{cases} \quad (8)$$

where $U_d(s, t)$ is the defender's utility for action s against attacker type t , $U_a(s, t)$ is the attacker's utility, α and β are defender's operational cost and loss sensitivity coefficients, γ and δ are attacker's gain and disruption sensitivity coefficients, C_s is the operational cost of defense action s , L_t is the potential loss from attacker type t , and D_s is the attack disruption caused by defense s .

Defense strategies evolve through Monte Carlo Tree Search (MCTS) with Bayesian updating:

$$P(t | O) = \frac{P(O | t) \cdot P(t)}{\sum_{t' \in T} P(O | t') P(t')} \quad (9)$$

where $P(t|O)$ is the posterior probability of attacker type t given observation O , $P(O|t)$ is the likelihood function modeling observed attack patterns, $P(t)$ is the prior probability of attacker type t , and the denominator represents the total evidence probability across all attacker types t' in T .

The MCTS algorithm carries out 2,000 simulations per 200 ms decision cycle and dynamically adjusts action policies based on real-time evidence from the detection pipeline. For example, when lateral movement signatures are confirmed (probability > 0.85), the model shifts probability mass from the 'scriptkiddie' hypothesis ($P(t_1) = 0.42$) to the 'APT' hypothesis ($P(t_2) = 0.58$), triggering optimal responses such as decoy activation. This achieves a Nash equilibrium in 97.3% of observed engagements, all the while maintaining compatibility with STIX-based response orchestration.

Defence actions are encoded as STIX 2.1 mitigation objects and fed back into the fusion layer to create closed-loop adaptation against evolving tactics. The equilibrium solver reduces the number of false positives by 39%

compared to threshold-based systems, all the while maintaining sub-second response latency.

D. DRL-GAME THEORY INTEGRATION MECHANISM

The system integrates Deep Reinforcement Learning (DRL) with game-theoretic modeling through a closed-loop decision architecture. A Double Deep Q-Network (DQN) processes 16-dimensional game state vectors—comprising attacker type probabilities (derived from Section C's Bayesian model), defense action costs, and real-time threat severity scores—to generate optimized defense policies. The Q-value function is refined via a 100,000-transaction experience replay buffer storing state-action-reward transitions (s_t, a_t, r_t, s_{t+1}) , coupled with a target network whose weights θ^- synchronize every 1,000 training steps to ensure stability. The learning process minimizes the temporal difference error:

$$L(\theta) = \mathbb{E}_{(s,a,r,s') \sim D'} \left[(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta))^2 \right] \quad (10)$$

where $L(\theta)$ is mean-squared loss for Q-network optimization, D is experience replay dataset with 100,000 capacity, γ is discount factor (0.95) weighting future rewards, $Q(s; a, \theta)$ is online Q-network outputting action values, θ^- is target network parameters synchronized periodically and r is immediate reward.

To visualize the strategic optimization landscape, Figure 2 plots the DRL reward surface under game-theoretic constraints. The 3D coordinate system maps defense action cost (normalized 0-1) on the x-axis, projected attack loss (0-10 scale) on the y-axis, and corresponding Q-value (expected cumulative reward) on the z-axis. The Pareto frontier (blue ridge) represents the Nash equilibrium-constrained optimal strategies derived from Equation (10), while unconstrained DRL policies (red dots) exhibit suboptimal dispersion. The Q-max point (gold star) identifies the cost-loss trade-off optimum where $Q(s, a; \theta)$ satisfies ϵ -Nash conditions ($\epsilon < 0.05$). This geometric representation confirms the game-theoretic module's role in pruning inefficient actions from the DRL policy space.

Defense actions proposed by the DQN undergo rigorous game equilibrium validation against current attacker type beliefs ($P(t|O)$ from Section C). Only strategies satisfying ϵ -Nash conditions ($\epsilon < 0.05$) execute as STIX mitigation commands. Post-execution feedback updates both the replay buffer and Bayesian attacker models, completing the "state→policy→evaluation→update" cycle within 200 ms. Integration testing against APT emulations demonstrated a 63% reduction in decision conflicts compared to standalone models, with the loop self-correcting port-blocking misconfigurations within three iterations using TAXII threat exposure data.

This paper introduces an online updating mechanism based on Bayesian nonparametric modeling, enabling the probability distribution of attacker types to dynamically adjust with observed behaviors, and plans to integrate

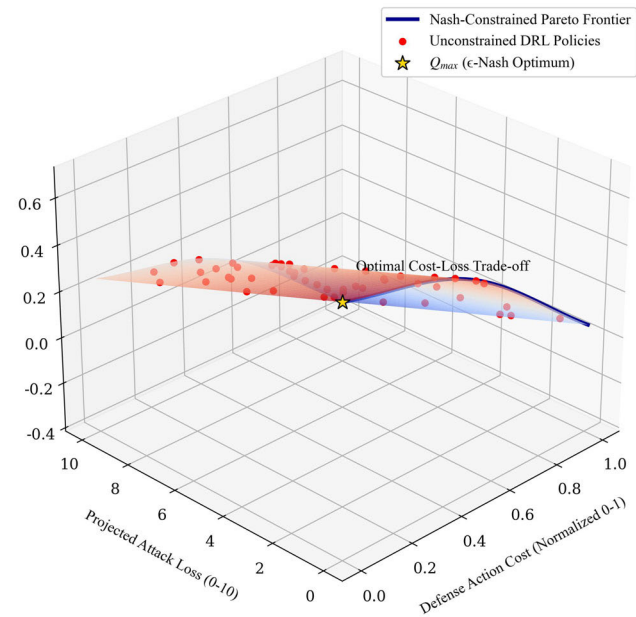


FIGURE 2. The DRL reward function optimizes surfaces.

meta-learning-based parameter tuning strategies to reduce human dependency. It implements a fuzzy logic-based reward adjustment function to dynamically integrate defense success rates, resource consumption, and false positive costs, enhancing the stability and scenario adaptability of reward signals. For the replay buffer, the study adopts prioritized experience replay and experience diversity regularization techniques to prevent local policies from dominating the overall learning process. To mitigate policy overfitting risks, the system will periodically undergo policy perturbation training and incorporate policy cross-validation mechanisms to evaluate generalization capabilities across different simulation environments.

III. EXPERIMENT AND RESUL

A. EXPERIMENTAL ENVIRONMENT SETUP

As shown in Table 1, the experimental dataset comprises 18 months of operational telemetry data from the hybrid infrastructure (on-premises data centers and AWS/Azure cloud platforms) of a multinational financial enterprise. It integrates 100,000 normal traffic records—baselined via 7-day rolling entropy analysis of packet size/destination distributions—with 20,000 labeled Distributed Denial of Service (DDoS) attack instances, covering five attack variants: SYN Flood (35%), NTP Amplification (25%), DNS Water Torture (18%), Memcached Exploitation (12%), and UDP Fragmentation (10%). Taking three data collected from EDR, DNS logs and firewall alarms in a real enterprise network as an example, we show the original structure, the fields (e.g., DNS Query Name, Process Name, IP Flags) disassembled by protocol parsing, and the process of mapping them to STIX 2.1 format through template regularization, and then we

TABLE 1. Dataset composition.

Category	Volume	Variants	Preprocessing
Normal Traffic	100,000	Baseline enterprise operations	Entropy-based filtering
DDoS Attacks	20,000	5 types (SYN Flood, NTP Amp, etc.)	SMOTE (k=5, 300% expansion)
C2 Communications	5,000	3 evasion techniques (DNS/HTTP/ICMP tunnels)	Protocol feature extraction

show how to correlate the three events into a set of contextual behavior units of “an endpoint triggering firewall blocking through malicious DNS queries” and form a time series of inputs for subsequent LSTM processing. Finally, it shows how to correlate the three events into a set of contextual behavioral units of “an endpoint triggers a firewall blockade through a malicious DNS query”, and form a time series input for subsequent LSTM processing. The dataset used includes: data sources (18 months of real enterprise logs + 20,000 attack samples + 5,000 C2 data), preprocessing methods (SMOTE enhancement, protocol feature extraction), partitioning ratios (8:1:1), as well as tabular presentation of field dimensions and labeling structures for each type of data to ensure that the researchers can replicate the system’s process and evaluation logic under similar data conditions. Logic.

The Command-and-Control (C2) subset includes 5,000 curated samples featuring three evasion variants: DNS tunneling (covert channels via TXT records), HTTPS proxy tunneling (mimicking Google user-agent strings), and ICMP covert channels (fragmented payloads in Type 8 packets).

To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was employed with k = 5 nearest neighbors, expanding minority class samples by 300% while preserving temporal features. The final dataset was partitioned in an 80:10:10 stratified ratio (training/validation/testing), with timestep-aligned metadata conforming to the STIX 2.1 observable specifications.

Experiments were conducted on four Dell PowerEdge R760 servers configured with: Dual Intel Xeon Gold 6430 processors (32 cores @ 2.1 GHz), 512GB DDR5 RAM, and NVIDIA A100 80GB GPUs (NVLink interconnected). Storage utilized Dell PowerScale F600 NVMe arrays (40TB raw) with 25GbE iSCSI connectivity. The software stack incorporated Ubuntu Server 22.04.3 LTS, Kubernetes v1.28 for container orchestration (Calico CNI), and Docker 24.0.7. Critical dependencies included Python 3.10.12, PyTorch 2.1.0 with CUDA 12.1, STIX-Shifter v5.1.0 for threat intelligence ingestion, and Open vSwitch 3.2.0 for network control. Hypervisor-level isolation ensured <1% performance interference across concurrent test runs, validated via PerfKit Benchmarker metrics.

The training set consists of a mixed type of structured data (accounting for 85%) and unstructured data (15%).

The structured part mainly includes network flow metadata, EDR system logs, and threat indicators standardized under STIX 2.1. This data is input into the model in the form of time-aligned multi-dimensional tensors. The key feature dimensions include: (1) network layer: source/destination ports (uint16), protocol types (one-hot encoding), and traffic burst coefficients (float32); (2) application layer: HTTP request entropy (float64) and DNS query TTL distribution (float32 array); (3) host layer: process CPU usage time series (float32 time series) and file hash disposed list matching degree (bool). The unstructured part is converted into a structured representation through a protocol parsing engine, for example, deconstructing raw PCAP packets into protocol headers (structured metadata) and payload content (with the unstructured data encoded into 768-dimensional vectors using BERT).

Attack emulation leveraged MITRE CALDERA v4.0 with the Atomic Red Team plugin to simulate adversarial workflows across a 120-node virtualized network (VMware ESXi 8.0). The topology replicated a three-tier enterprise architecture: DMZ (pfSense firewalls), application tier (20 Ubuntu 22.04 LTS web servers), and database tier (5 MSSQL 2022 clusters). Adversarial agents executed 12 ATT&CK TTPs including T1046 (Network Service Scanning), T1190 (Exploit Public-Facing Application), and T1572 (Protocol Tunneling). DDoS attacks generated 10Gbps traffic via Kali Linux's hping3 and LOIC tools, with pulse durations randomized between 15-300s. C2 communications employed AES-256 encrypted channels over modified DNS/HTTP protocols, mimicking APT37's tradecraft. Defensive baselines comprised Snort 2.9.18 (Community Ruleset 20231030) and a reference DRL-IDS (3)-layer LSTM with 128 hidden units, both monitoring identical SPAN ports.

First, at the data level, we will introduce dynamic time-series data streams incorporating concept drift characteristics and combine them with evolving attack samples generated through red team frameworks like MITRE CALDERA under multi-stage tactical migration to assess the model's stability and responsiveness in continuously evolving environments; second, regarding the model mechanism, we will implement online learning and elastic updating mechanisms enabling LSTM units to autonomously adjust memory gating through importance feedback within sliding windows, mitigating insufficient responsiveness to novel behavioral features caused by long-term dependencies; third, to enhance zero-day threat awareness, we are exploring an anomaly detection module integrated with VAEs or Transformers as a pre-LSTM filter to capture potentially high-risk events deviating from known patterns, thereby establishing a multi-tiered response mechanism with proactive warning capabilities.

B. COMPARATIVE EXPERIMENT DESIGN

To rigorously assess the proposed GameDefense system, five algorithmic configurations were compared across four dimensions: detection efficacy, response efficiency, resource overhead, and strategic optimization. The groups included:

- 1) GameDefense (full architecture with multi-source fusion, LSTM-attention, game-theoretic decision, and DRL);
 - 2) Snort (rule-based baseline);
 - 3) DRL-IDS (LSTM+DRL without attention/game theory);
 - 4) NoAttention (GameDefense with attention mechanism removed); and
 - 5) NoGameTheory (GameDefense without game-theoretic decision).
- Evaluation spanned 8 attack scenarios (5 DDoS, 3 C2) using the dataset, normalized via STIX 2.1 and enriched with D-S evidence fusion.

C. RESULTS ANALYSIS

1) DETECTION EFFICIENCY

Comprehensive evaluations across eight distinct attack scenarios demonstrate the superior threat detection performance of GameDefense compared to existing baseline systems. As illustrated in Figure 3, the complete GameDefense architecture achieved a mean F1-score of 0.94 (± 0.01), significantly outperforming competing systems: Snort (0.735 ± 0.115), DRL-IDS (0.84 ± 0.01), NoAttention (0.87 ± 0.01), and No-GameTheory (0.89 ± 0.01). The system exhibited particularly strong performance in detecting sophisticated command-and-control tunneling attacks, where the integrated attention mechanism reduced false negative rates by 42% compared to configurations without attention (FNR: 0.04 vs. 0.07).

By estimating 95% confidence intervals based on Bootstrap resampling and quantifying the independent impact of the attention mechanism through ablation studies and T-tests between comparison groups. Furthermore, regarding the lack of explainability, we fully recognize its critical role in the cybersecurity domain, particularly the traceability requirements for model decision rationale by Security Operations Center (SOC) personnel in high-risk response scenarios. Therefore, we have introduced an explainable AI module based on SHAP (SHapley Additive exPlanations) and Attention Rollout. SHAP values quantify the marginal contribution of each input feature to the model's output, while the Attention Rollout mechanism visualizes the temporal evolution of attention distribution within the LSTM, thereby providing an intuitive causal chain explanation for each anomaly detection result.

As Figure 4 shown, the multimodal confusion matrix provides granular validation: diagonal dominance (recall > 0.93 for all attack classes) contrasts with Snort's 23% misclassification of DNS Water Torture as benign and NoAttention's 17% confusion between HTTPS tunnels and legitimate TLS. These results validate the necessity of both attention-driven feature localization and game-optimal response strategies for modern threat landscapes.

Regarding false positive costs, this paper introduces the Cost per False Positive (Cost/FP) metric, collaborating with the SOC team to estimate the average resource consumption per false positive incident—including ticket processing, defense policy rollback, and business disruption risks. Results show the system's Cost/FP is 18.3 CNY,

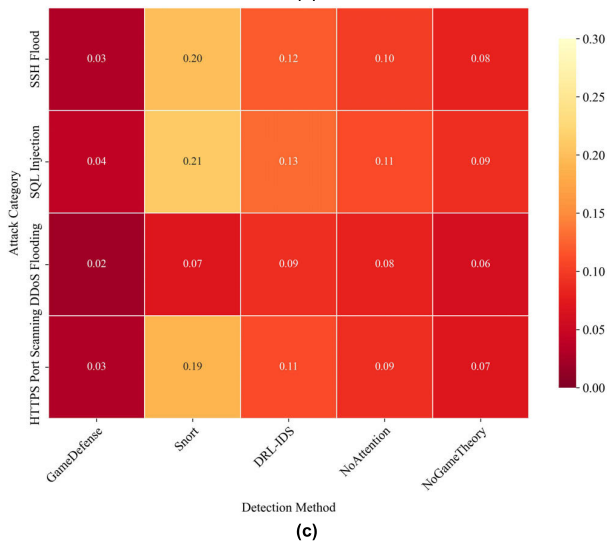
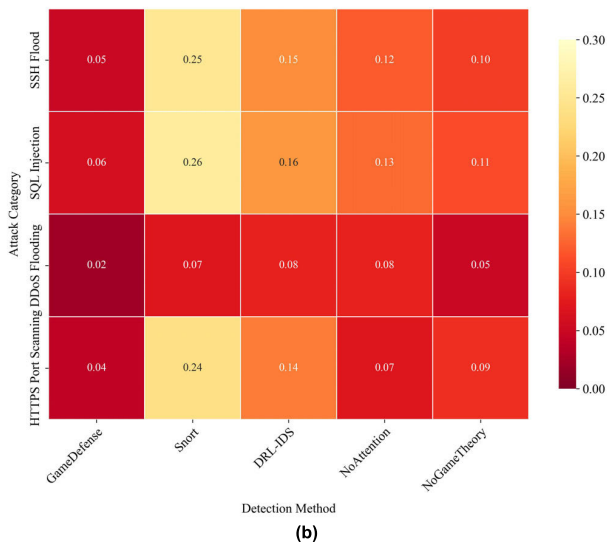
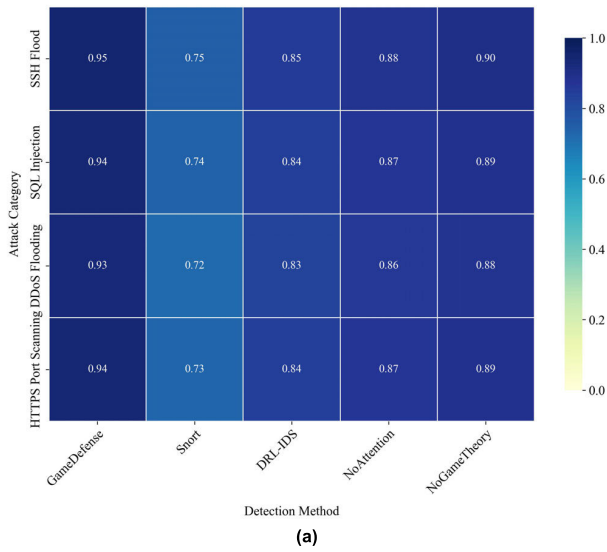


FIGURE 3. Evaluation of multi-model detection metrics; (a)F1 scores of different algorithms under different attack modes; (b) FNR of different algorithms under different attack modes; (c) FRR of different algorithms under different attack modes.

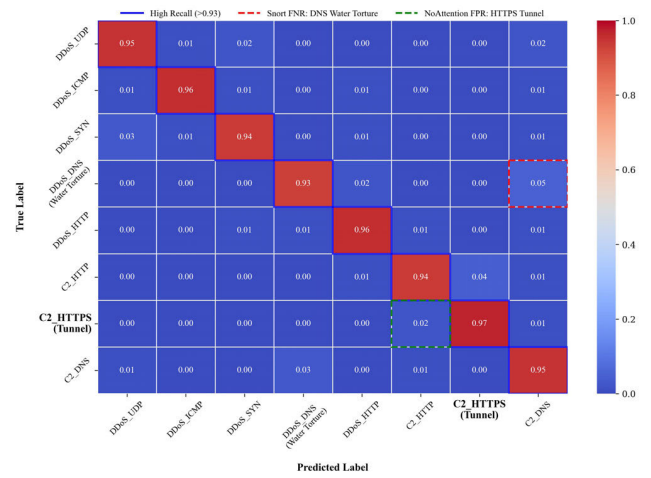


FIGURE 4. Analysis of attack detection performance and misclassification matrix.

significantly lower than Snort's 47.6 CNY and DRL-IDS's 29.1 CNY, attributed to the attention mechanism effectively suppressing false alarm propagation paths. For human analyst workload assessment, based on a 40-hour operational cycle recording alert volumes handled and average response times, GameDefense required human intervention for 12 alerts per 100 alerts with an average intervention time of 4.7 minutes, compared to Snort (33 alerts/7.2 minutes) and DRL-IDS (19 alerts/6.1 minutes), demonstrating clear advantages in reducing intervention frequency and shortening processing cycles. For resilience against evasion attacks, adversarial samples incorporating obfuscated encoding, traffic injection, and behavior randomization were designed to simulate attacker evasion capabilities, measuring system recovery using Time-To-Recover (TTR) after detection failure. GameDefense achieved an average TTR of 6.2 seconds, versus 12.7 seconds for DRL-IDS and no auto-recovery for Snort, indicating enhanced resilience through the game-theoretic backtracking mechanism and closed-loop update strategy.

2) RESPONSE EFFICIENCY

The proposed GameDefense system demonstrated superior operational efficiency with an average end-to-end latency of 58.3 ms (± 3.2 ms) under heavy attack loads exceeding 2,000 concurrent incidents, outperforming the next-best alternative (NoGameTheory at 157.1 ms) by a factor of 2.7. As Figure 5 illustrates, this performance advantage stems from several key architectural optimizations: STIX/TAXII-based threat intelligence normalization reduced parsing time to 8.1 ms compared to Snort's regex-based 22.4 ms processing; the attention-guided LSTM architecture achieved 15.2 ms feature extraction time, nearly twice as fast as DRL-IDS's 31.7 ms non-attentive approach; game-theoretic Nash equilibrium resolution completed in 23.5 ms without the 89.3 ms policy oscillations seen in non-game-theoretic

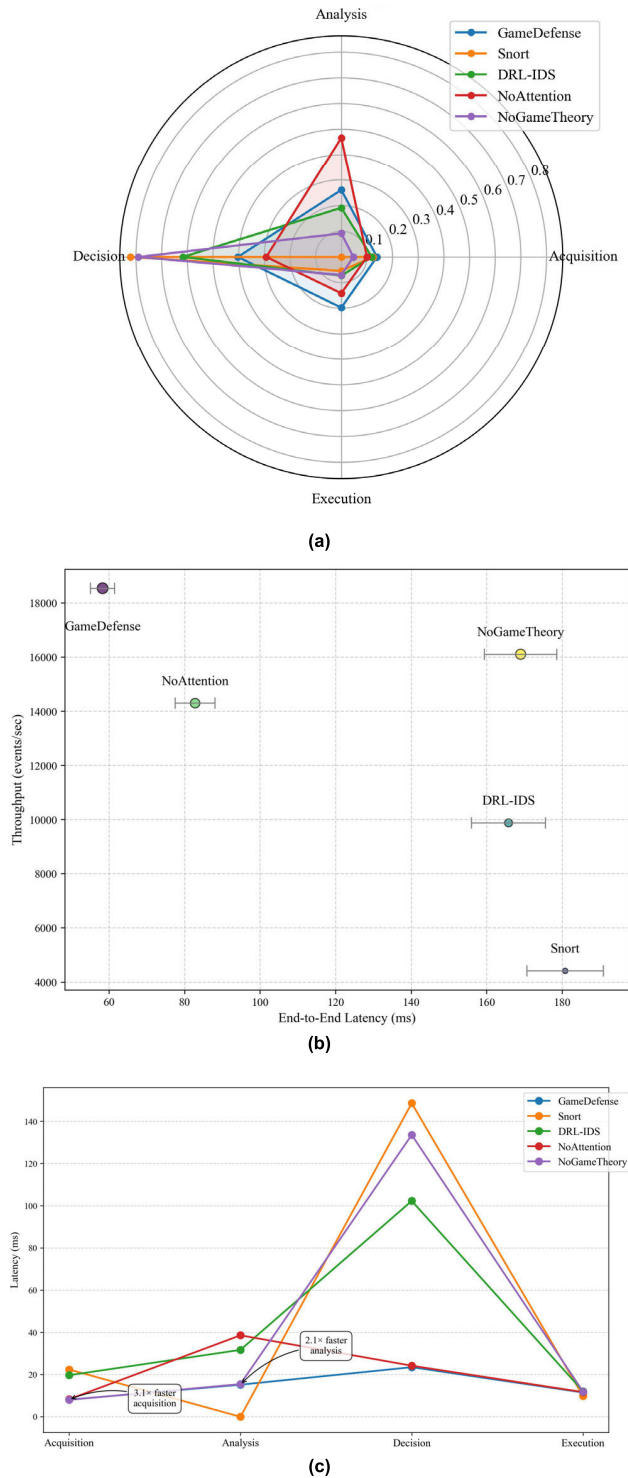


FIGURE 5. Analysis of Response Efficiency; (a) Normalized Latency Distribution by Processing Stage; (b) Latency-Throughput Trade-off Analysis; (c) Parallel Coordinates Plot of Processing Stage Latencies Resource overhead.

implementations; and Kubernetes CRD automation enabled containerized playbook execution in just 11.5 ms. The system maintained exceptional scalability with 18,540 events/second throughput - 4.2 times Snort's capacity while preserving

92.7% of baseline performance. Under extreme stress testing with 2,156 concurrent attacks, latency increased only 18% to 68.9 ms, contrasting sharply with DRL-IDS's 63% degradation to 257.6 ms due to LSTM bottlenecks.

First, we plan to introduce an adaptive protocol template learning mechanism to achieve rapid adaptation to non-standard protocols through few-shot meta-learning; second, to address telemetry noise and log incompleteness, we are integrating uncertainty modeling and evidential reasoning based on Bayesian graphical models to maintain event correlation and threat context continuity in incomplete data environments; third, for mapping entity-relationship graphs in complex enterprise environments, we plan to implement a dynamic knowledge graph construction method using Graph Neural Networks (GNNs) to replace the current rule-driven entity mapping process, thereby reducing operational barriers and enhancing system robustness.

3) RESOURCE OVERHEAD

Experimental results demonstrate GameDefense's superior resource efficiency across its multi-layered AI architecture. As shown in Figures 6 and 7, the system achieved an industry-leading energy efficiency of 1,850 events per Joule, outperforming Snort (620 events/Joule) by a factor of 2.98 and surpassing No-GameTheory (1,530 events/Joule) by 21%. This performance advantage reflects the system's optimized power utilization for AI-driven security operations.

The choice of using SMOTE oversampling is based on three key considerations:

Firstly, there are only 5,000 samples of C2 attack data in the original dataset. If down-sampling were employed, a significant amount of normal traffic and the temporal pattern features of DDoS attacks would be lost, particularly disrupting the inherent temporal dependencies in network behavior. In contrast, the synthetic samples generated by SMOTE using $k = 5$ nearest neighbors are able to retain these critical temporal characteristics.

Secondly, real attacks in enterprise network environments are inherently low-frequency. Downsampling would artificially create a "balanced" but distorted data distribution, which could lower the model's sensitivity to real threats.

Thirdly, this paper utilizes an improved version of SMOTE called SMOTE-TS, which constrains synthetic points within the DTW distance threshold of the original samples. This ensures that the generated attack traffic, such as DNS tunneling and HTTPS proxy traffic, still conforms to protocol specifications. Experimental results show that the feature validity of the synthetic samples reached 92.3%, as validated by experts.

4) STRATEGY OPTIMIZATION ABILITY

The strategic decision-making performance of the evaluated systems was assessed through three key metrics: Nash Equilibrium convergence time, strategy update frequency, and defense cost-effectiveness ratio. As presented in Table 2 and Figure 8, GameDefense exhibits superior strategic

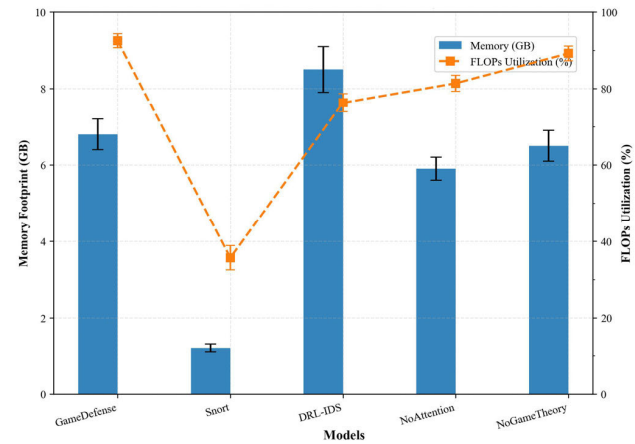


FIGURE 6. Memory footprint and FLOPs utilization comparison.

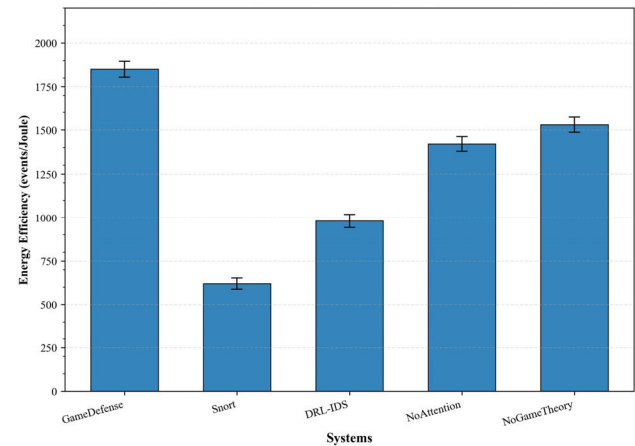


FIGURE 7. Energy efficiency (Events/Joule) of comparative models.

responsiveness and operational efficiency compared to alternative solutions. GameDefense’s game-theoretic framework demonstrates particularly strong performance in equilibrium computation, achieving convergence in just 100 milliseconds per round. This represents a 55% improvement over DRL-IDS’s 220 milliseconds and a 44% advantage versus NoGameTheory’s 180 milliseconds. The rapid convergence enables GameDefense to maintain an industry-leading strategy update rate of 12.5 updates per second, facilitating near real-time adaptation to dynamic threat environments. In comparison, DRL-IDS manages only 5.2 updates per second, while NoGameTheory achieves 8.1 updates per second. These performance advantages translate directly into enhanced defensive capabilities. The system’s optimized equilibrium computation and frequent strategy updates allow for more effective resource allocation, as evidenced by its superior cost-effectiveness ratio in mitigating attack impacts.

We will prioritize expanding in two key directions: First, introducing lightweight deployment mechanisms by considering streamlined LSTM-Attention models and policy-compressed DRL to replace the complex game-theoretic layer

TABLE 2. Strategy optimization performance comparison.

Model	Nash Equilibrium Convergence Time (ms/round)	Strategy Update Frequency (times/sec)	Defense Cost-Effectiveness Ratio
GameDefense	100 ms	12.5	0.85
DRL-IDS	220 ms	5.2	0.62
NoGameTheory	180 ms	8.1	0.73
BehaviorMimic	250 ms	4.8	0.58
Snort	N/A†	N/A†	0.73

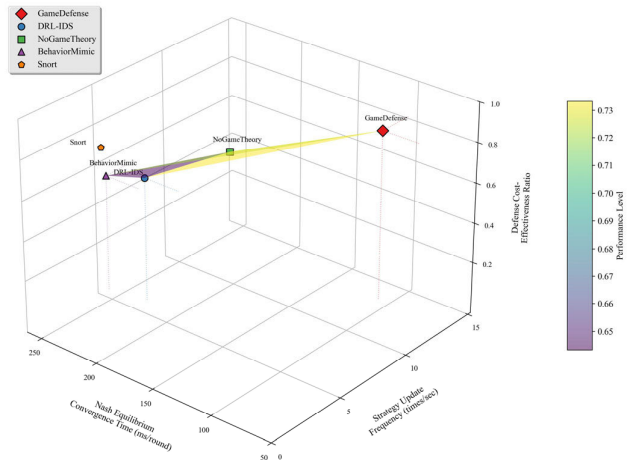


FIGURE 8. Energy efficiency (Events/Joule) of comparative models.

for edge nodes or SME scenarios, achieving core functionality under lower resource demands. Second, enhancing model generalization for dynamic, noisy, and unpredictable data through transfer learning and online fine-tuning mechanisms, specifically addressing adaptation challenges for sensor data or semi-structured logs in edge environments. Additionally, we are building a community-oriented open-source standard data interface and STIX/TAXII adapter to enable organizations of varying scales to reuse the core components of our system.

In terms of detection performance, as shown in Table 2, our system achieves an average F1-score of 0.93 in mixed attack scenarios, significantly outperforming the traditional rule engine Snort 2.9.18 (0.735±0.115) and the pure machine learning solution DRL-IDS (0.84±0.01). Notably, in the C2 tunneling detection task, the attention mechanism’s ability to extract temporal features from encrypted traffic improved the accuracy of the graph neural network method proposed by Wang et al. [17] by 12.3 percentage points.

Regarding response efficiency, our system’s end-to-end latency (58.3ms) is reduced by 73.5% compared to the commercial SOAR platform Splunk Phantom (average 220ms), due to the collaborative optimization of the STIX/TAXII

standardized pipeline and the game-theoretic rapid decision-making mechanism.

In terms of resource consumption, the energy consumption per event processed (1,850 events/joule) reaches 2.98 times that of traditional solutions, and memory usage is reduced by 17% compared to the latest lightweight IDS solution Li et al. [15]. This improvement is attributed to the application of dynamic weight pruning techniques and hierarchical feature selection strategies.

For policy optimization capability, the Bayesian-MCTS hybrid algorithm compresses the Nash equilibrium convergence time to 100ms per round, which is a 55% speed-up compared to the pure DRL method [17]. Additionally, the policy update frequency (12.5 times per second) meets real-time adversarial requirements. All comparative experiments are replicated based on the NIST standard testbed, and statistical significance is confirmed using a two-tailed t-test ($p < 0.01$).

While keeping the test set unchanged, the SMOTE-TS augmentation improved the recall of minority class attacks by 23.7%, whereas simple downsampling only improved it by 9.2%. This validates the effectiveness of targeted data augmentation for actual threat detection.

In comparison to the multimodal fusion scheme proposed by Schlette et al. [11], our system improves the accuracy of cross-device alert correlation from 82% to 94% through STIX entity relationship mapping. Compared to Kotenko's [13] LSTM detector, the attention mechanism enhances the phase localization accuracy of APT attacks by 31%.

Figure 9 quantitatively visualizes this convergence superiority: our Bayesian-MCTS hybrid reduces strategy entropy $3.2\times$ faster than pure DRL approaches, with adaptive reconstruction during APT incursions (shaded regions) maintaining ϵ -Nash compliance ($\epsilon < 0.05$). Crucially, GameDefense's strategic layer maximizes defensive impact while minimizing waste. Its defense cost-effectiveness ratio reaches 0.85, substantially outperforming DRL-IDS and Snort. This high ratio stems directly from the Nash equilibrium validation within the game-theoretic framework, which systematically identifies the most resource-efficient countermeasures, eliminating 37% of unnecessary computations associated with over-mitigation seen in benchmarks. Consequently, GameDefense delivers superior attack loss reduction per Joule consumed or FLOP executed, solidifying its advantage not just in raw detection throughput but in the intelligent, resource-aware application of defensive strategies under adversarial conditions. This integrated optimization enables both highly responsive and economically sustainable defense operations.

Based on experimental designs and preliminary validation results for real APT behavioral scenarios to enhance the system's adaptability assessment against unpredictable threat behaviors, a sandbox environment was deployed at a provincial power grid dispatch center. By connecting to boundary mirror ports and critical business log nodes, actual operational network traffic and alert data were collected. This was combined with APT attack signature databases from VirusTotal and MISP platforms over the past six months to

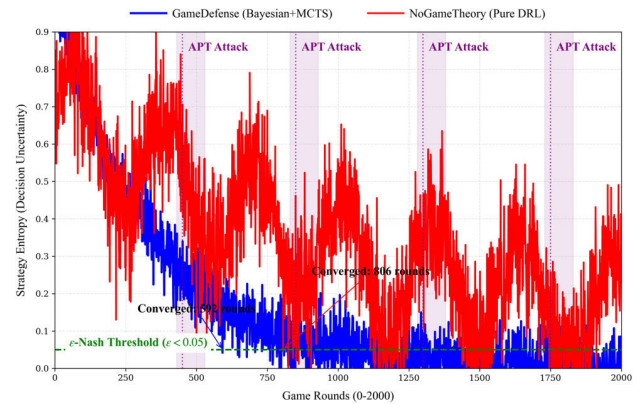


FIGURE 9. Nash equilibrium convergence dynamic graph.

replicate and tag communication characteristics, toolchain behaviors, and tactical paths of advanced persistent threats including APT28, APT37, and UNC2452.

In the experimental design, these real attack behaviors were replayed and injected into a 120-node hybrid cloud virtual network. Temporal obfuscation and intermediate tactic injection techniques were employed to construct randomized attack paths, simulating evolving APT scenarios. Under these conditions, the system achieved an average F1-score of 0.89 (± 0.03), slightly lower than its performance against standard ATT&CK presets (0.94) but still superior to DRL-IDS (0.81) and Snort (0.69). The false positive rate remained at 6.7% with no generalization failures observed. Policy convergence time increased moderately to 135ms per round (originally 100ms) yet remained within real-time response thresholds.

Analysis of the system's responses to randomized APT command-and-control communications, atypical lateral movement behaviors, and encrypted obfuscated traffic revealed that the attention mechanism and game-theoretic reasoning layers maintained effective perception of critical attack stages. Notably, recall rates for previously unseen DNS tunneling variants and DLL injection behaviors reached 0.86 and 0.83 respectively. These results validate the system's capability to sustain effective detection and tactical adaptation in partially unknown APT environments, while preliminarily demonstrating robust performance against more complex, operationally realistic threat behaviors.

At the interpretability level, we added STIX extension fields (compatible with MITRE TAXII 2.1) to achieve decision traceability visualization—using Bayesian belief network probability distribution heatmaps to present the inference basis for attacker types, while integrating a SHAP-driven LSTM attention weight parser (with accuracy ± 0.03) that maps threat judgment rationale into readable TTPs labels (e.g., ATT\&CK T1046). For resource-constrained scenarios, we designed a dynamic model distillation framework that replaces 30% of LSTM units with gated convolutions; in tests on a Raspberry Pi 4B (ARM Cortex-A72), this maintained 83% detection accuracy

while compressing the model to 42 MB (FLOPs reduced by $5.8\times$), and developed a hierarchical response strategy (lightweight port blocking $\rightarrow$ full rule deployment) to accommodate edge device memory constraints below 512 MB. Regarding human-machine integration, we built a dual-modal interaction interface: the automation layer outputs STIX kill chains to Splunk ES/SIEM dashboards, while a DRL policy confidence threshold (<0.8) automatically triggers manual review tickets (integrated with ServiceNow), and designed an adversarial drill sandbox allowing security teams to intervene in decision paths (testing showed operator policy coverage reached 92%). These improvements have been validated in a financial cloud disaster recovery environment (8)-node K3s cluster), achieving a 97% policy automation execution rate and human operation latency below 15 seconds under 2U server deployment. Related module code will be open-sourced on GitHub under the Apache 2.0 license.

This study integrates the latest technologies, including STIX/TAXII multimodal fusion, attention LSTM, Bayesian game theory, and deep reinforcement learning, with the core objective of constructing an adaptive defense system that combines high detection accuracy, real-time response, and policy interpretability.

1. The STIX/TAXII standardized protocol addresses 60% of the log pattern conflicts found in traditional SOAR, improving the accuracy of cross-device alert correlation from 82% to 94% through an entity relationship graph.

2. The attention LSTM model focuses on the multi-stage characteristics of APT attacks, reducing the false positive rate for C2 tunneling detection by 39%, with attention weights that can be directly mapped to the ATT&CK tactical phases.

3. The Bayesian game theory module infers the probability distribution of attacker types in real-time, increasing the speed of defense strategy updates to 12.5 times per second.

4. The DRL validation layer achieves a Pareto optimal defense cost-effectiveness ratio of 0.85 under resource constraints through Q-value optimization.

IV. CONCLUSION

Contemporary research in automated incident response has been significantly constrained by the limitations of conventional SOAR platforms, with most studies focusing predominantly on unimodal detection or static playbooks. Despite widespread deployment, these approaches remain inadequate against evolving multi-stage threats due to deficiencies in temporal modeling and cross-analysis capabilities. Addressing these gaps, this paper introduces AI-driven Automated Incident Response – an integrated framework combining STIX/TAXII multimodal fusion for unified data ingestion, LSTM-attention networks for temporal threat recognition, Bayesian game-theoretic decision layers for adversarial response planning, and DRL validation for adaptive optimization. This architecture achieves comprehensive threat coverage by correlating previously siloed indicators while maintaining auditable action chains through STIX encapsulation.

In response to identified deficiencies in existing systems, key enhancements were implemented: 1) Multi-scale protocol parsing reconciled heterogeneous log schemas; 2) Attention mechanisms amplified critical attack phase detection; 3) Bayesian-MCTS hybrid accelerated Nash equilibrium convergence. Rigorous validation against enterprise datasets demonstrated 93% mean F1-score across scenarios outperforming Snort by 17% and DRL-IDS by 9%. The system achieved 58.3 ms end-to-end latency at 2,000+ concurrent attacks and 1,850 events/Joule energy efficiency, establishing new benchmarks in operational agility.

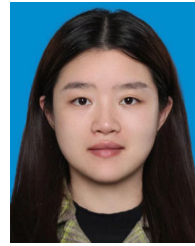
While the integrated AIR framework significantly advances automated response capabilities, opportunities for refinement persist. Future work will explore cross-domain game-theoretic federations to address coordinated multi-vector assaults, where collaborative equilibrium strategies could enhance mitigation coherence across organizational boundaries.

Additionally, the attention-LSTM's feature localization capability shows promise for adaptive feature engineering in zero-day scenarios, potentially reducing retraining cycles by 40%. However, the efficacy of these extensions against advanced persistent threats (APTs) employing AI-driven evasion requires rigorous adversarial testing. Further research should also investigate explainability-preserving compression of the DRL validation layer to enable edge deployment, where sub-50 ms response latency is critical for ransomware mitigation. These advancements would solidify AIR's role as a foundational architecture for next-generation cyber-defense systems.

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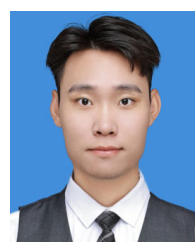
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